rational numbers using integers.

13th October Lecture-16

Recap:

1) Sets, operation on sets
2) Sequences
3) functions, composition of functions, injective, surjective & bijective
4) permutation function
5) Pigeon hole principle
6) Binary relations.

Definition [Equivalence classes]:

Let A be a non-empty set and R be an equivalence relation over A, i.e., $R \subseteq A \times A$. Let $x \in A$. Then, the equivalence class of x with respect to R is defined as follows:

$$[x]_R := \{a \in A : (a, x) \in R\}.$$

The set of all equivalence classes of R is denoted as

$$A/R := \{[x]_R : x \in A\}.$$

- Note that $A/R \subseteq \text{pow}(A)$.

Examples:

i) Consider the 'equal to' relation $(=)$ on $\mathbb{Z}$. We have seen that it is an equivalence relation. Then, for every $a \in \mathbb{Z}$,

$$[a]_{=} = \{a\}.$$

Equivalence class — $[a]_{=}$ — relation

ii) Let R be a relation on $\mathbb{Z} \times \mathbb{Z}^*$ defined by $((a,b),(c,d)) \in R$ if and only if $ad = bc$. We have seen that R is an equivalence relation. Let $(a,b) \in \mathbb{Z} \times \mathbb{Z}^*$ be arbitrary. Then,

$$[(a,b)]_R = \{(c,d) \in \mathbb{Z} \times \mathbb{Z}^* \mid \frac{a}{b} = \frac{c}{d}\}.$$

$ad = bc$

It means that the elements of $[(a,b)]_R$ can be viewed as all rational numbers that are equal to $\frac{a}{b}$.

Remark: This equivalence relation plays a very important role in constructing rational numbers using integers.

Lemma 4.6.1 [Properties of equivalence classes]: Let A be a non-empty set & R be an equivalence relation on A. Then,

a) for every $x \in A$, $x \in [x]_R$.

b) for every $x, y \in A$, $(x,y) \in R$ if and only if $\boxed{[x]_R = [y]_R.}$

c) for every $x, y \in A$, $(x,y) \notin R$ if and only if $[x]_R \cap [y]_R = \emptyset$.

Proof: a) Let $x \in A$ be arbitrary. Then, as R is an equivalence relation, it is also reflexive, which implies $(x,x) \in R$. Hence, $x \in [x]_R$.

b) Let $x, y \in A$. Suppose $(x,y) \in R$. As R is an equivalence relation, it is symmetric and transitive. Let $a \in [x]_R$ be arbitrary. Then, $(a,x) \in R$.

We know $(x,y) \in R$. Now, by the transitivity of R, $(a,y) \in R$. This implies $a \in [y]_R$, which implies $[x]_R \subseteq [y]_R$. Similarly, we can show $[y]_R \subseteq [x]_R$. Hence, $[x]_R = [y]_R$.

Now, suppose $[x]_R = [y]_R$. By the first point, $x \in [x]_R$ and $y \in [y]_R$. As $[x]_R = [y]_R$, we get that $x \in [y]_R$, which implies $(x,y) \in R$.

iii) Let $x, y \in A$ be arbitrary. Suppose $(x,y) \notin R$. Let $a \in [x]_R$ & $a \in [y]_R$. Then, $(a,x), (a,y) \in R$.

By symmetry of R, $(x,a) \in R$ & by transitivity of R, $(x,y) \notin R$. This is not true & hence $[x]_R \cap [y]_R = \phi$.

Now, suppose $[x]_R \cap [y]_R = \phi$.

We want to show that $(x,y) \notin R$.

Let's do it by contraposition.

Let $(x,y) \in R$. Then, it follows that $x,y \in [x]_R \cap [y]_R$ & hence $[x]_R \cap [y]_R \neq \phi$.

■

Definition [Partition]: Let A be a non-empty set & P be a collection of some subsets of A. Then, the

collection P is called a partition of A if and only if the following properties hold:

✓ a) for every $X \in P$, $X \neq \emptyset$.

✓ b) for every $X, Y \in P$, either $X = Y$ or $X \cap Y = \emptyset$.

✓ c) for every $a \in A$, there exists an $X \in P$ s.t. $a \in X$.

Example:

a) Let O & E be sets of odd numbers & even numbers. Let $P = \{O, E\}$. Then, P is a partition of $\mathbb{Z}$.

b) Let A be any non-empty

& $B \subsetneq A$. Then, $P := \{B, A-B\}$ is a partition of A. (HW)

The following claim and lemma relate partitions & equivalence relations.

Claim 4.6.2: Let A be a non-empty set & R be an equivalence relation on A. Then, the set A/R is a partition on A.

Proof: Homework.

Lemma 4.6.3: Let A be a non-empty set and P be a partition of A. Let R be a relation defined on

A as follows: for every $a, b \in A$, $(a,b) \in R$ if and only if there exists $X \in P$ such that $a, b \in X$. Then, R is an equivalence relation and $A/R = P$.

Proof: a) R is reflexive: HW.

b) R is symmetric: HW.

c) R is transitive: Let $a, b, c \in A$, such that $(a,b), (b,c) \in R$. This means there exist $X, Y \in P$ such that $a, b \in X$ and $b, c \in Y$. As P is a partition, either $X \cap Y = \phi$ or $X = Y$. We know $b \in X \cap Y$, hence $X = Y$. Thus,

$a, b, c \in X$. Then, by definition of R, $(a,c) \in R$. Thus, R is transitive.

Thus, R is an equivalence relation.

Now, we want to show that $A/R = P$. Let $a \in A$ be arbitrary. Then, there exists $X \in P$ such that $a \in X$. As P is a partition of A, we get that such an X exists and it is unique. Thus,

$$[a]_R = \{b \in X\} = X.$$

This implies $[a]_R \in P$ for every $a \in A$, hence $A/R \subseteq P$. Now, let $X \in P$ be arbitrary. As P is

a partition, $X \neq \phi$. Let $a \in X$ be arbitrary. Then, we know by the definition of R that

$$X = [a]_R.$$

This implies $X \in A/R$ & $P \subseteq A/R$.

Hence, $P = A/R$. ∎

4.7 Countable and uncountable sets

Recall that a set X is said to be finite if and only if either $X = \phi$ or elements of X can be labelled as $a_1, \ldots, a_n$ for some $n \in \mathbb{N}$. For example, the set of students in Delhi, the empty set, etc.

Definition [Cardinality of a finite set]: Let X be a finite set. Then, the cardinality of X, denoted $|X|$, is defined as the number of elements in X.

For example $|\{a \in \mathbb{N} \mid a \leq 10\}| = 11$.

Claim 4.7.1 [Closure of finite sets under set operations]: Let X, Y be finite sets. Then, $X \cup Y$, $X \cap Y$, $X - Y$, $Y - X$, $X \times Y$, $Y \times X$ are also finite.

Proof: It is not difficult to argue the following.

a) $|X \cup Y| \leq |X| + |Y|$.

b) $|X \cap Y| \leq \min\{|X|, |Y|\}$.

c) $|X - Y| \leq |X|$.

d) $|Y - X| \leq |Y|$.

e) $|X \times Y| = |Y \times X| = |X| \cdot |Y|$.

Homework: Prove these using the definitions of various set operations.

15th October Lecture-17

Recap:

1) Sets
2) Sequences
3) functions, permutations
4) Pigeonhole principle
5) Relations & partitions
6) finite sets.

- **Infinite sets** : These are the sets which are not finite.

Definition [Countably infinite set]:
A set X is said to be countably

infinite if and only if there exists a bijective function

$F : \mathbb{N} \longrightarrow X$.

E.g. 1) $\mathbb{N}$ is countably infinite.

2) (HW): Let X be a finite set. Prove that it is not countably infinite.

Claim 4.7.1 : The set of integers $\mathbb{Z}$ is countably infinite.

Proof: To prove this, we have to show that there

exists a bijective function

$F: \mathbb{N} \to \mathbb{Z}$.

Consider the following function

$F: \mathbb{N} \to \mathbb{Z}$ such that for every $a \in \mathbb{N}$,

$$f(a) = \begin{cases} \frac{a}{2} & \text{if } a \text{ is even} \\ -\frac{(a+1)}{2} & \text{if } a \text{ is odd} \end{cases}$$

We claim that f is bijective.

a) <u>F is injective</u>: We have to show that for every $a_1, a_2 \in \mathbb{N}$, $a_1 \neq a_2$, $f(a_1) \neq f(a_2)$.

We prove this by cases. (natural numbers)

i) Both a_1 & a_2 are even:

Then, $a_1 = 2b_1$ & $a_2 = 2b_2$

for some $b_1, b_2 \in \mathbb{N}$.

$a_1 \neq a_2$

HW $\Rightarrow$ $f(a_1) \neq f(a_2)$

$\|$ $\|$

$b_1 \neq b_2$

2) Both a_1 & a_2 are odd:

By definition, $a_1 = 2b_1 + 1$

& $a_2 = 2b_2 + 1$ for $b_1, b_2 \in \mathbb{N}$

In this case, $f(a_1) = -\frac{(2b+1+1)}{2}$

$= -(b_1 + 1)$.

Similarly, $f(a_2) = -(b_2 + 1)$.

$a_1 \neq a_2$

HW

$\Rightarrow F(a_1) \neq f(a_2)$.

3) Exactly one of a_1 is even & the other is odd:

Without loss of generality, suppose a_1 is even & a_2 is odd. Then, by definition, there exist $b_1, b_2 \in \mathbb{N}$ such that

$$a_1 = 2b_1, \quad a_2 = 2b_2 + 1.$$

$$F(a_1) = b_1, \quad F(a_2) = \frac{-(2b_2+1+1)}{2}$$

$$= -(b_2+1)$$

we are given $a_1 \neq a_2$ &
we want to show $f(a_1) \neq f(a_2)$
, the same as $b_1 \neq -(b_2+1)$.

Homework.

ii) f is surjective: Homework.

□

Homework: Prove that the sets of even [odd] integers is countably infinite.

Claim 4.7.2: The set $\mathbb{N} \times \mathbb{N}$ is countably infinite.

Proof: We have to give a

bijective function from $\mathbb{N}$ to $\mathbb{N}\times\mathbb{N}$.

$$f: \mathbb{N} \longrightarrow \mathbb{N}\times\mathbb{N}$$
$$i \longmapsto (i, i).$$

$$f(i) = \begin{cases} (i, \frac{i}{2}) & i \text{ even} \\ (i, \frac{i+1}{2}) & i \text{ odd} \end{cases}$$

$$F(i) = (i, \mathbb{N}) \in \mathbb{N}\times\mathbb{N}$$

$$f(i) = (i, i^2)$$

$$f(i) = (j, i) \quad j \neq i$$

$$\mathbb{N}\times\mathbb{N} = \{(a,b) \mid a, b \in \mathbb{N}\}.$$

$$\mathbb{N}\times\mathbb{N} = \{(0,0), (0,1), (0,2), \dots, (1,0), (1,1), \dots\}$$

infinite

infinite

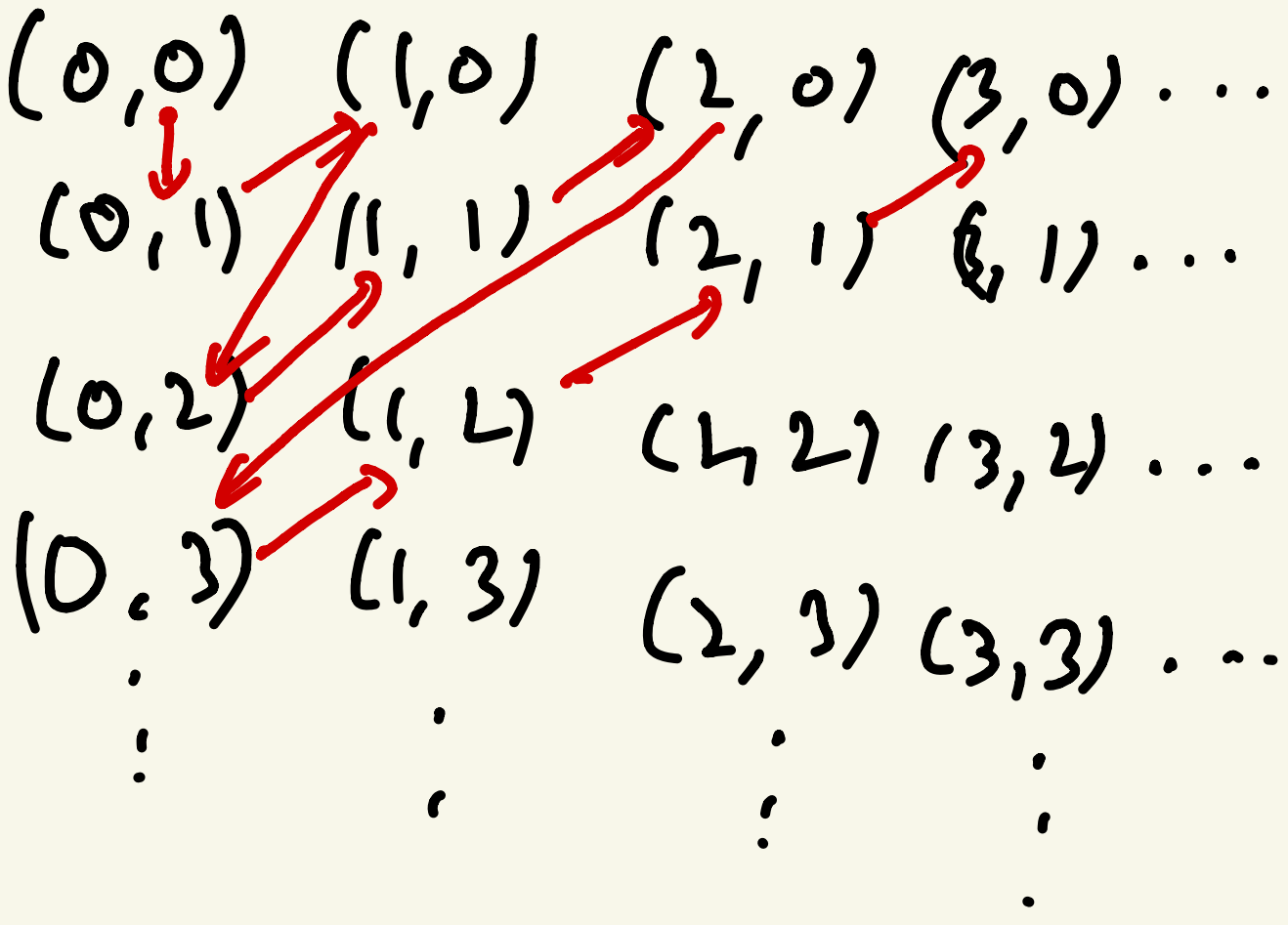
$$\mathbb{N}\times\mathbb{N} = \{(0,0),(0,1),(1,0),(0,2),(1,1),$$
$$(2,0),(0,3),(1,2),(2,1),$$
$$(3,0),\ldots\}$$

We consider the ordered representation of $\mathbb{N}\times\mathbb{N}$ given above

$$\mathbb{N}\times\mathbb{N} = [\underbrace{(0,0)}_{0},\underbrace{(0,1)}_{1},\underbrace{(1,0)}_{2},\underbrace{(0,2)}_{3},\underbrace{(1,1)}_{4},\underbrace{(2,0)}_{5},\ldots] \quad -(1)$$

The idea behind this ordered view of $\mathbb{N}\times\mathbb{N}$ is that we first list all pairs in $\mathbb{N}\times\mathbb{N}$ whose sum is 0, then all pairs whose sum is 1, & so on.

Among pairs having the same sum, we follow a structured ordering as follows: for every such pair (a, b), we write these pairs in the increasing order of values of a.



This gives us the following function $F : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$

$i \longmapsto (x_{i+1}, y_{i+1})$.

for every $i \in \mathbb{N}$, $f(i)$ is $\| (x_{i+1}, y_{i+1})$ the $(i+1)$th element in the ordered set $\mathbb{N} \times \mathbb{N}$ given in Equation ①.

i) F is surjective: Let (a,y) be arbitrary element of $\mathbb{N} \times \mathbb{N}$. Let (a,b) be located at j-th position in Eq ①. Then, by definition $f(j-1) = j$.

2) f is injective: Let $i, j \in \mathbb{N}$.
Suppose $i \neq j$. Then,
we want to prove that
$f(i) \neq f(j)$. As $i \neq j$,
we have two possibilities,
either $i < j$ or $i > j$.
Let $i < j$. Then, we know
from the ordered view
of $\mathbb{N} \times \mathbb{N}$ given in Equation ①
that there exist $(x_1, y_1), (x_2, y_2)$
$\in \mathbb{N} \times \mathbb{N}$ such that
$f(i) = (x_1, y_1)$ and
$f(j) = (x_2, y_2)$

We want to show that $(x_1, y_1) \neq (x_2, y_2)$. Either $x_1 \neq x_2$ or $y_1 \neq y_2$.

As $i < j$, we have $i+1 < j+1$. Thus, we either have $x_1 + y_1 < x_2 + y_2$ or we have $x_1 + y_1 = x_2 + y_2$ and $x_1 < x_2$. In both the cases, $(x_1, y_1) \neq (x_2, y_2)$.

Hence, F is injective.

□

Remark: Using a similar argument, we can show that $\mathbb{Q}$ is

also countably infinite.

29th October Lecture-18

Recap:

a) finite sets : closed under every set operation
b) Countably infinite sets
c) Examples

Definition [Countable set]:
A set is said to be countable if either it is finite or countably infinite.

Remark: We can enumerate the elements of a countable set, i.e., say which is the first element, the second element and so on.

One may ask if every infinite set is countably infinite. In the late 19th century, Georg Cantor showed that there are infinite sets which are not countable.

Definition [Uncountable sets]: A set is called uncountable if it is not countable.

We will see some examples of uncountable sets. We need the following definition to give the first example of an uncountable set.

Definition [Characteristic function of a set]: Let A be a countable set with its elements enumerated as $a_1, a_2, a_3, \ldots$. Let $S \subseteq A$. Then, the characteristic function of S, denoted χ_S (chi), is defined as follows: for every $a_i \in A$,

$$\chi_S : A \to \{0,1\}$$

$$\chi_S(a_i) := \begin{cases} 1 & \text{if } a_i \in S \\ 0 & \text{otherwise} \end{cases}$$

for example, let $A = \{1, 2, 3\}$ & $S = \{2, 3\}$. Then, $\chi_S(1) = 0$, $\chi_S(2) = 1$ & $\chi_S(3) = 1$.

A useful representation of χ_S:
As A is a countable set, we can represent the characteristic function χ_S as follows: χ_S is a sequence whose entries are labelled by every element of A in order $a_1, a_2, a_3, \ldots$, and the entry labelled by a_i is

equal to 1 if and only if $a_i \in S$.

$S \subseteq A$, $a_1, a_3, a_4 \in S$, $a_2 \notin S$,

for example,

$$\chi_S = \begin{matrix} a_1 & a_2 & a_3 & a_4 & \cdots \\ (0 & 1 & 0 & 0 & \cdots) \end{matrix}$$

This representation of χ_S will be very useful to show the existence of uncountable sets.

Theorem 4.7.3: The power set of natural numbers, denoted pow(ℕ) is uncountable.

Proof: As ℕ is infinite, clearly pow(ℕ) is also infinite.

Assume for the contradiction that $\mathrm{pow}(\mathbb{N})$ is countably infinite. We know that

$$\mathrm{pow}(\mathbb{N}) := \{B \subseteq \mathbb{N}\}.$$

This means that there exists a bijective function

$$F : \mathbb{N} \longrightarrow \mathrm{pow}(\mathbb{N}).$$

We will eventually show that there exists $A \subseteq \mathbb{N}$ such that for every $i \in \mathbb{N}$, $f(i) \neq A$. In other words, we will show that f is not surjective.

As per our assumption $pow(\mathbb{N})$ is countably infinite. Hence, we can enumerate the elements of $pow(\mathbb{N})$ with respect to the function f. Suppose

$S_0 := f(0), S_1 := f(1), \ldots, S_i := f(i), \ldots,$

where every $S_i \in pow(\mathbb{N})$, equivalently, $S_i \subseteq \mathbb{N}$.

Consider the following table

	0	1	2	3	...
χ_{S_0}	a_{00}	a_{01}	a_{02}	a_{03}	...
χ_{S_1}	a_{10}	a_{11}	a_{12}	a_{13}	...
χ_{S_2}	a_{20}	a_{21}	a_{22}	a_{23}	...
χ_{S_3}	a_{30}	a_{31}	a_{32}	a_{33}	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

where the $(i,j)^{th}$ entry, a_{ij} is defined as

$$a_{ij} := \chi_{S_i}(j) \begin{cases} 0, & \text{if } j \notin S_i \\ 1, & j \in S_i \end{cases}$$

Every entry of this table is either 0 or 1.

Let us define the set $D \subseteq \mathbb{N}$ such that that characteristic function of D, denoted χ_D is defined as follows:

$\chi_D(0) := 1 - a_{00}$, $\chi_D(1) := 1 - a_{11}$, $\chi_D(2) := 1 - a_{22}$, $\ldots$, $\chi_D(i) := 1 - a_{ii}$

$\ldots$

Clearly, $D \in pow(\mathbb{N})$.

we will show that χ_D is not present the table above we know that pow($\mathbb{N}$) is countably infinite, i.e., we can list all the subsets of $\mathbb{N}$ as $S_0, S_1, S_2, \ldots$. As $D \subseteq \mathbb{N}$, it should be present in this list, i.e., $D = S_i$ for some $i \in \mathbb{N}$. That means that the i-th row of the table is χ_D. But by definition, $\chi_{S_i}(i) = a_{ii}$ & as $S_i = D$, $\chi_{S_i}(i) = \chi_D(i)$. But, we have defined

$$\chi_D(i) = 1 - a_{ii}.$$

Hence, we get $a_{ii} = 1 - a_{ii}$

$$\Rightarrow \boxed{1 = 2 \cdot a_{ii}.}$$

Since $a_{ii} \in \{0, 1\}$, this is not possible. Hence we get a contradiction & $\wp(\mathbb{N})$ is uncountable.

Remark: This proof technique is known as Cantor's diagonalization argument.

Theorem 4.7.4: Let X & Y be two sets.

a) If X, Y are countable then $X \cup Y$ & $X \cap Y$ is also countable.

b) If X is countable then every subset $A \subseteq X$ is also countable.

contraposition

c) If there exists $A \subseteq X$ such that A is uncountable then X is uncountable.

d) If X and Y are

countably infinite then there exists a bijective function $F: X \to Y$.
(equivalently, a bijective function $G: Y \to X$).

Proof: a) Will be in the assignment.

b) Given that X is countable, i.e., X is either finite or X is countably infinite.
Let $A \subseteq X$ be arbitrary.
If A is finite then there is nothing to prove.

Assume A is infinite & X is countably infinite. Then, there exists a bijective function

$$F: \mathbb{N} \to X.$$

Suppose $i_1, i_2, i_3, \ldots$ be an infinite sequence of natural numbers such that $f(i_1) =: a_1$, $f(i_2) =: a_2$, $\ldots$, $f(i_j) =: a_j, \ldots$

and $a_1, a_2, a_3, \ldots, a_j, \ldots$ are all the elements of A.

We want to show that there exists a bijective function from $\mathbb{N}$ to A.

Consider

$$G : \mathbb{N} \longrightarrow A$$

$$\forall j \in \mathbb{N}, \quad j \longmapsto a_j = F(j)$$

in other words $G(j) = a_j$.

H.W.: Prove that G is a bijective function.

c) Follows from b).

d) X & Y are countably infinite. There exists

$$H : \mathbb{N} \to X, \quad K : \mathbb{N} \to Y$$

such that both H & K are bijective.

Let $a_0 := H(0), a_1 := H(1), \ldots$

and $b_0 := K(0), b_1 := K(1), \ldots$

We want to give a bijective function from X to Y.

$$F : X \longrightarrow Y$$

for every $i \in \mathbb{N}$, $F(a_i) := b_i$

H.W: Prove that F is bijective.

∎

Theorem 4.7.5: The interval $[0,1]$ is uncountable, where $[0,1] := \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

Proof: We know that every element in $[0,1]$ can be written as $0.d_1 d_2 d_3 \ldots$, where every digit $d_i \in \{0,1,\ldots,9\}$ & there can be countably infinite many such digits. Consider the set

$$A := \{0.d_1 d_2 d_3 \ldots \mid \text{every } d_i \in \{0,1\}\}.$$

We will prove that A is uncountable. Then, by point c of Theorem 4.7.4, we will get that $[0,1]$ is also uncountable.

Assume for the contradiction that A is countably infinite. Then, there exists a bijective function $F: \mathbb{N} \to A$.

Then, we can enumerate all the elements of A as follows: $F(0) := 0.d_{00}d_{01}d_{02}\ldots$, $F(1) := 0.d_{10}d_{11}d_{12}, \ldots, \ldots$ & so on.

We know that every $d_{ij} \in \{0,1\}$.

Now we will show that there exists $x \in A$ such that $x \neq f(i)$ for every $i \in \mathbb{N}$.

We will use the Cantor's diagonalization argument.

Define x as follows

$$x := 0.\underbrace{(1-d_{00})}_{0\text{-th digit}}\underbrace{(1-d_{11})}_{1\text{st digit}}(1-d_{22})\cdots$$

Clearly $x \in A$.

Is x present in $f(0), f(1), \ldots$?

HW.: Answer it.

□

Corollary 4.7.6: $\mathbb{R}$ & $\mathbb{C}$ are uncountable sets.

3rd Nov Lecture-19

Recap

a) Logic
- Propositional
- Predicate

b) Proofs
- Modus Ponens
- Direct
- Contraposition
- Biconditional
- Counter example
- Contradiction

c) Set theory
- sets & operations
- functions
- Pigeonhole Principle
- Relations
- finite & infinite sets

5. Proof by induction

This is a very important proof methodology. This technique is often used to prove that some property holds for all natural numbers greater than or equal to a specific $n_0 \in \mathbb{N}$.

The template of proof by induction is as follows:

Let $P(n)$ be a proposition about natural numbers n. We want to prove that $P(n)$ is true for every $n \in \mathbb{N}$, $n \geq n_0$, where $n_0 \in \mathbb{N}$ is some fixed

natural number. Then, this proof technique has the following structure.

P(n) is true $\forall n \geq n_0$

a) **Base case**: We first **prove** that $P(n_0)$ is true.

b) **Induction hypothesis**: Let $n \in \mathbb{N}$, $n > n_0$ be arbitrary. We **assume** that $P(n)$ is true.

c) **Induction step**: In this step, using the induction hypothesis, we **prove** that $P(n+1)$ is true.

- **Why induction is a valid proof technique?:** In this method, we first prove $P(n_0)$ is true, then assume $P(n)$ is true & then prove $P(n) \Rightarrow P(n+1)$ is true, where $n > n_0$, $n \in \mathbb{N}$ is arbitrary. Let's see why assuming $P(n)$ is true does not create any problem.

Here, we are proving

$$P(n_0) \Rightarrow P(n_0+1) \Rightarrow \cdots \Rightarrow P(n) \Rightarrow P(n+1) \Rightarrow \cdots$$

is true. To do this, we prove $P(n_0)$ & $P(n) \Rightarrow P(n+1)$ are

true. Then, by Modus Ponens, $P(n_0+1), \dots, P(n)$ must be true. Hence, we can safely assume that $P(n)$ is true in the induction hypothesis.

Notation: Let $a_1, \dots, a_n \in \mathbb{N}$ for some $n \in \mathbb{N}$. Then, $\sum_{i=1}^{n} a_i$ means $a_1 + a_2 + \dots + a_n$. In general, let $i, j \in \{1, \dots, n\}$ $i < n$. $\sum_{k=i}^{j} a_k = a_i + a_{i+1} + \dots + a_j$

Claim 5.1: for every $n \in \mathbb{N}$, $n \geq 1$,

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

Proof: Let $P(n)$ be the proposition $\sum_{i=1}^{n} i = \frac{n \cdot (n+1)}{2}$.

We want to prove that $P(n)$ is true for every $n \geq 1$ (no).

1) **Base case**: We want to prove that $P(1)$ is true, i.e.

$$1 = \frac{1(1+1)}{2}$$

Trivially, $P(1)$ is true.

2) **Induction hypothesis**:

Let $n \in \mathbb{N}$ be arbitrary satisfying $n > 1$. Assume

that $P(n)$ is true, i.e.,

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

3) Induction step: We want to prove that $P(n+1)$ is true, i.e.,

$$\sum_{i=1}^{n+1} i = \frac{(n+1)(n+2)}{2}.$$

Note that

$$\sum_{i=1}^{n+1} i = \boxed{\sum_{i=1}^{n} i} + (n+1)$$

By the induction hypothesis, we know $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

$$\sum_{i=1}^{n+1} i = \frac{n(n+1)}{2} + (n+1)$$

$$= (n+1)\left[\frac{n}{2} + 1\right]$$

(By distributivity)

$$= (n+1)\left(\frac{n+2}{2}\right)$$

$$\sum_{i=1}^{n+1} = \frac{(n+1)(n+2)}{2}$$

□

Claim 5.2 [Sum of geometric progression]:

Let $r \in \mathbb{N}$ such that $r \neq 1$. Then, for every $n \in \mathbb{N}$, $n \geq 1$,

$$\sum_{i=0}^{n} x^i = \frac{x^{n+1}-1}{x-1}.$$

Proof: Let $P(n)$ be the proposition $\sum_{i=0}^{n} x^i = \frac{x^{n+1}-1}{x-1}$ for $n \in \mathbb{N}$. We prove that $P(n)$ is true for every $n \geq \underbrace{1}_{n_0}$ by **induction on n.**

1) **Base case:** By definition, $P(1)$ is $\sum_{i=0}^{1} x^i = \frac{x^2-1}{x-1}$

$x^0 + x^1 = 1 + x$ $(\because x^0 = 1)$

$$1+x = \frac{(x+1)(x-1)}{(x-1)\in} = \frac{x^2-1}{x-1}$$

Hence, $P(1)$ is true.

2) Induction hypothesis:

Let n be arbitary.

We assume that $P(n)$ is true, i.e.,

$$\sum_{i=0}^{n} x^i = \frac{x^{n+1}-1}{x-1}.$$

3) Induction step: We want to prove that $P(n+1)$ is true, i.e.,

$$\sum_{i=0}^{n+1} x^i = \frac{x^{n+2}-1}{x-1}.$$

we have

$$\sum_{i=0}^{n+1} x^i = \boxed{\sum_{i=0}^{n} x^i} + x^{n+1}$$

$$= \frac{x^{n+1}-1}{x-1} + x^{n+1}$$

(By induction hypothesis)

$$= \frac{x^{n+1}}{x-1} - \frac{1}{x-1} + x^{n+1}$$

$$= \frac{z^{n+1}}{z-1} + z^{n+1} - \frac{1}{z-1}$$

$$= z^{n+1}\left[\frac{1}{z-1} + 1\right] - \frac{1}{z-1}$$

[By distributivity]

$$= z^{n+1}\left[\frac{z}{z-1}\right] - \frac{1}{z-1}$$

$$= \frac{z^{n+2}}{z-1} - \frac{1}{z-1}$$

$$= \frac{z^{n+2} - 1}{z-1}$$

□

Homework: for every $n \in \mathbb{N}$, $n \geq 1$

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Claim 5.3: For every $n \in \mathbb{N}$, $n \geq 10$, $n^3 \leq 2^n$.

Proof: Let $P(n)$ be the proposition $n^3 \leq 2^n$. We prove by induction that $P(n)$ is true for every

$n \geq \boxed{10}$.
n_0

i) Base case: We have to show that $P(10)$ is true,

i.e., $10^3 \leq 2^{10}$.

$1000 \qquad 1024$

Hence, $P(10)$ is true.

2) Induction hypothesis:

Let $n \in \mathbb{N}$, $n > 10$ be arbitrary. Assume that $P(n)$ is true, i.e.,

$n^3 \leq 2^n$.

3) Induction step: we want to prove that $P(n+1)$ is true, i.e.,

$$(n+1)^3 \leq 2^{(n+1)}.$$

We know that

$$(n+1)^3 = n^3 + 1 + 3n^2 + 3n$$

To show that $(n+1)^3 \leq 2^{n+1}$ we have to prove that

$$\boxed{3n^2 + 3n + 1 \leq n^3}$$

$3n^2+3n+1 \leq n^3$

$\Leftrightarrow \quad 3n^2+3n \leq n^3-1$

$\Leftrightarrow \quad 3n(n+1) \leq n^3-1$

$\Leftrightarrow \quad n+1 \leq \frac{1}{3n}(n^3-1)$

$\Leftrightarrow \quad n+1 \leq \frac{1}{3}\left(n^2-\frac{1}{n}\right)$

$\Leftrightarrow \quad n \leq \frac{1}{3}\left(n^2-\frac{1}{n}\right)-1$

$\Leftrightarrow \quad 1 \leq \frac{1}{3n}\left(n^2-\frac{1}{n}\right)-\frac{1}{n}$

$\Leftrightarrow \quad 1 \leq \frac{1}{3}\left(n-\frac{1}{n^2}\right)-\frac{1}{n}$

$\Leftrightarrow \quad 1 \leq \underbrace{\frac{n}{3}}_{>1} - \underbrace{\left(\frac{1}{n}+\frac{1}{3n^2}\right)}_{<1}$.

H.W: Prove that for $n \geq 10$, $n \in \mathbb{N}$, $\frac{1}{n} + \frac{1}{3n^2} < 1$.

As $n > 10$, $\frac{n}{3} > 3$.

we get that the RHS is greater than 1.

Hence, $3n^2 + 3n + 1 \leq n^3$.

$\Rightarrow (n+1)^3 \leq n^3 + n^3$

By the induction hypothesis

$(n+1)^3 \leq 2^{n+1}$.

Remark: Let $d \in \mathbb{N}$. Then

there exists $n_0 \in \mathbb{N}$ s.t.
for every $n \geq n_0$,
$$n^d \leq 2^n.$$

Homework: for every $n \in \mathbb{N}$, $n \geq 2$, prove that
$$2^n < (n+1)!,$$
where $(n+1)! := (n+1)\, n \cdot (n-1) \cdots 2 \cdot 1$

Claim 5.4 [Bernoulli's inequality]
Let $x \in \mathbb{R}$ such that $x > -1$.
Then, for every $n \in \mathbb{N}$, $n \geq 1$,

$$(1+x)^n \geq 1+nx.$$

Proof: Let $P(n)$ be the proposition $(1+x)^n \geq 1+nx$.

We prove the claim by induction on n for every $n \geq 1$.

i) Base case: $P(1)$ is trivially true.

2) Induction hypothesis: Let $n \in \mathbb{N}$, $n > 1$ be arbitrary. Assume that

$P(n)$ is true, i.e.,

$$(1+x)^n \geq 1+nx.$$

3) Induction step: We want to prove that $P(n+1)$ is true, i.e., $(1+x)^{n+1} \geq 1+(n+1)x$

We have

$$(1+x)^{n+1} = (1+x)^n \cdot (1+x)$$

By the induction hypothesis, as

$$(1+x)^n \geq 1+nx$$

$$\Rightarrow (1+x)^{n+1} \geq (1+nx)(1+x)$$

[as $x > -1$, $1+x > 0$ &

hence the inequality doesn't flip).

$$(1+nx)(1+x) = 1+nx+x+nx^2$$
$$= \underline{1+(n+1)x} + \boxed{nx^2}$$

As x is a real number & n is a non-zero natural number, we have $nx^2 \geq 0$.

$$1+(n+1)x+nx^2 \geq 1+(n+1)x.$$

We had

$$(1+x)^{n+1} \geq 1+(n+1)x+nx^2$$

Hence, $(1+x)^{n+1} \geq 1+(n+1)x.$

■

Claim 5.5: Let X and Y be finite non-empty sets & $f: X \to Y$ be a function. Suppose $|X| > |Y|$. Then, f is not injective.

Proof: As X & Y are non-empty finite sets & $|X| > |Y|$, clearly, $|X| \geq 2$.

We want to show that there exist $x_1, x_2 \in X, x_1 \neq x_2$ such that $f(x_1) = f(x_2)$.

we prove the claim by induction on $|X|$ for $|X| \geq 2$.

Let $|X| = n$. Then, $P(n)$ is the following proposition:

If $|X| = n$, $|X| > |Y|$, $f: X \to Y$ is an arbitrary function, then there exist $x_1, x_2 \in X$, $x_1 \neq x_2$ s.t. $f(x_1) = f(x_2)$.

1) Base case: Let $n = 2$. Then, as $|X| > |Y|$, $|Y| = 1$.

Let $X = \{x_1, x_2\}$ & $Y = \{y_1\}$.

We know that f is a function, hence

$f = \{(x_1, y), (x_2, y)\}$.

Hence, f is not injective in this case.

2) Induction hypothesis:

Let $n \in \mathbb{N}$, $n > 2$ be arbitrary. Assume $P(n)$ is true.

3) Induction step: We want to prove that $P(n+1)$ is true. We have $|X| = n+1$ & $|Y| \leq n$. & $f : X \to Y$ is an

arbitrary function.

Let $X = \{x_1, \ldots, x_{n+1}\}$.

Then, there are two cases:

Case 1: There exists an $i \in \{2, \ldots, n+1\}$ s.t. $f(x_1) = f(x_i)$.

we are done here.

Case 2: for every $i \in \{2, \ldots, n+1\}$ $f(x_i) \neq f(x_1)$.

Let $X' := X - \{x_1\} \Rightarrow |X'| = n$

and $Y' := Y - \{f(x_1)\}$.

$\Rightarrow |Y'| \leq n-1$

Let $f' : X' \to Y'$ be a function defined as follows: for every $i \in \{2, \ldots, n+1\}$

$$f'(x_i) := f(x_i).$$

As f is a function, f' is also a function. By induction hypothesis, there exist $x_i, x_j \in X'$ such that $x_i \neq x_j$ but

$$f'(x_i) = f'(x_j)$$

As $f'(x_i) = f(x_i)$,

we get that $f(x_i) = f(x_j)$.

Hence f is not injective

■